

Started on	Saturday, 31 January 2026, 10:00 AM
State	Finished
Completed on	Saturday, 31 January 2026, 10:55 AM
Time taken	55 mins 20 secs
Grade	14.00 out of 15.00 (93.33%)

Question 1

Correct

Mark 1.00 out of 1.00

Given: A is 3×3 ,

$$(B^T - 2I)^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad |3A| = 27, \quad \text{tr}(3A) = 15.$$

Find $|A^{-1}|$

- ☐ A. **27**
- ☒ B. **1** ✓
- ☐ C. **$\frac{1}{3}$**
- ☐ D. **$\frac{1}{27}$**

Your answer is correct.

The correct answer is: **1**

Question 2

Correct

Mark 1.00 out of 1.00

Given: A is 3×3 ,

$$(B^T - 2I)^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad |3A| = 27, \quad \text{tr}(3A) = 15.$$

Find B and B^2

- ☐ A. $B = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, B^2 = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- ☒ B. $B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}, B^2 = \begin{bmatrix} 25 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ ✓
- ☐ C. $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, B^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}$
- ☐ D. $B = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \end{bmatrix}, B^2 = \begin{bmatrix} 49 & 0 & 0 \\ 0 & 36 & 0 \\ 0 & 0 & 25 \end{bmatrix}$

Your answer is correct.

The correct answer is: $B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}, B^2 = \begin{bmatrix} 25 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 9 \end{bmatrix}$

Question 3

Correct

Mark 1.00 out of 1.00

Given: A is 3×3 ,

$$(B^T - 2I)^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad |3A| = 27, \quad \text{tr}(3A) = 15.$$

Find $|2AB|$

- ☐ A. 60
- ☐ B. 240
- ☒ C. 480 ✓
- ☐ D. 960

Your answer is correct.

The correct answer is: 480

Question 4

Correct

Mark 1.00 out of 1.00

Given: A is 3×3 ,

$$(B^T - 2I)^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad |3A| = 27, \quad \text{tr}(3A) = 15.$$

Find $\text{tr}(2A + 3B)$

- ☐ A. 36
- ☒ B. 46 ✓
- ☐ C. 56
- ☐ D. 26

Your answer is correct.

The correct answer is: 46

Question 5

Correct

Mark 1.00 out of 1.00

Encode the message “*Help him now*” using the given table and the matrix, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

= 0	I = 9	R = 18
A = 1	J = 10	S = 19
B = 2	K = 11	T = 20
C = 3	L = 12	U = 21
D = 4	M = 13	V = 22
E = 5	N = 14	W = 23
F = 6	O = 15	X = 24
G = 7	P = 16	Y = 25
H = 8	Q = 17	Z = 26

Choose the correct encoded sequence:

- ☒ A. 13, 5, 28, 16, 8, 8, 22, 13, 14, 14, 38, 23 ✓
- ☐ B. 13, 28, 8, 22, 14, 38, 5, 16, 8, 13, 14, 23
- ☐ C. 8, 5, 12, 16, 0, 8, 9, 13, 0, 14, 15, 23
- ☐ D. 5, 13, 16, 28, 8, 8, 13, 22, 14, 14, 23, 38

Your answer is correct.

The correct answer is: 13, 5, 28, 16, 8, 8, 22, 13, 14, 14, 38, 23

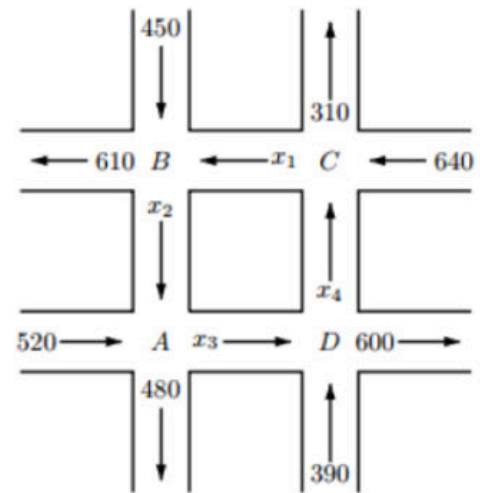
Question 6

Correct

Mark 1.00 out of 1.00

In an AI-based traffic digital twin, the unknown internal one-way flows are x_1, x_2, x_3, x_4 . Using flow conservation at each junction (Input = Output), The governing linear system is:

Which of the following systems matches the conservation equations for the network?



- ☒ A. $\begin{cases} x_2 + 520 = x_3 + 480 \\ x_1 + 450 = x_2 + 610 \\ x_4 + 640 = x_1 + 310 \\ x_3 + 390 = x_4 + 600 \end{cases}$ ✓
☐ B. $\begin{cases} x_3 + 520 = x_2 + 480 \\ x_2 + 450 = x_1 + 610 \\ x_1 + 640 = x_4 + 310 \\ x_4 + 390 = x_3 + 600 \end{cases}$
☐ C. $\begin{cases} x_2 + 480 = x_3 + 520 \\ x_1 + 610 = x_2 + 450 \\ x_4 + 310 = x_1 + 640 \\ x_3 + 600 = x_4 + 390 \end{cases}$
☐ D. $\begin{cases} x_2 - 520 = x_3 - 480 \\ x_1 - 450 = x_2 - 610 \\ x_4 - 640 = x_1 - 310 \\ x_3 - 390 = x_4 - 600 \end{cases}$

Your answer is correct.

The correct answer is: $\begin{cases} x_2 + 520 = x_3 + 480 \\ x_1 + 450 = x_2 + 610 \\ x_4 + 640 = x_1 + 310 \\ x_3 + 390 = x_4 + 600 \end{cases}$

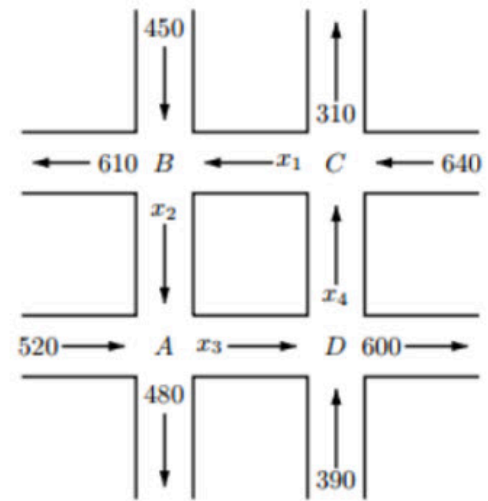
Question 7

Correct

Mark 1.00 out of 1.00

In an AI-based traffic digital twin, the unknown internal one-way flows are x_1, x_2, x_3, x_4 . Using flow conservation at each junction (Input = Output), The governing linear system is:

For the same traffic model, which statement is correct?



- ☐ A. The system has a unique solution:
 $(x_1, x_2, x_3, x_4) = (330, 170, 210, 0)$
- ☒ B. The system has infinitely many solutions: ✓
 $(x_1, x_2, x_3, x_4) = (t + 330, t + 170, t + 210, t), t \geq 0$
- ☐ C. The system has no solution because the equations are inconsistent.
- ☐ D. The system has infinitely many solutions:
 $(x_1, x_2, x_3, x_4) = (t, t + 330, t + 170, t + 210), t \geq 0$

Your answer is correct.

The correct answer is:

The system has infinitely many solutions:

$$(x_1, x_2, x_3, x_4) = (t + 330, t + 170, t + 210, t), t \geq 0$$

Question 8

Correct

Mark 1.00 out of 1.00

In the traffic digital twin model, the internal flows satisfy the general solution:

$$(x_1, x_2, x_3, x_4) = (t + 330, t + 170, t + 210, t)$$

where t is a free parameter.

Assume two practical AI constraints from sensors/road capacity:

- Nonnegativity: $x_i \geq 0$ for all i
- Capacity constraint: $x_1 \leq 500$

Which range of t makes the solution physically feasible?

- ☐ A. $0 \leq t \leq 500$
- ☐ B. $-330 \leq t \leq 170$
- ☒ C. $0 \leq t \leq 170$ ✓
- ☐ D. $t \geq 0$ only

Your answer is correct.

The correct answer is: $0 \leq t \leq 170$

Question 9

Incorrect

Mark 0.00 out of 1.00

In AI/Data Science, eigenvectors define “special directions” that are scaled (not rotated) by a linear transform A .

Let A be an $n \times n$ matrix and $v \neq 0$ be an eigenvector of A with eigenvalue λ , i.e. $Av = \lambda v$.

Which statement is always true?

- ☐ A. v is also an eigenvector of A^2 with eigenvalue λ^2 .
- ☐ B. v is always an eigenvector of A^{-1} with eigenvalue λ .
- ☒ C. If v is an eigenvector of A , then v must be an eigenvector of A^T . ✗
- ☐ D. If $Av = \lambda v$, then v is also an eigenvector of $A + I$, with eigenvalue λ .

Your answer is incorrect.

The correct answer is: v is also an eigenvector of A^2 with eigenvalue λ^2 .

Question 10

Correct

Mark 1.00 out of 1.00

In signal processing and dynamical models used in AI, matrix functions like $\cos(A)$ are defined using power series and can be computed efficiently if A is diagonalizable.

Assume A is diagonalizable:

$$A = PDP^{-1}, \quad D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Which expression is correct for $\cos(A)$?

- ☒ A. $\cos(A) = P \cos(D) P^{-1}$, where $\cos(D) = \text{diag}(\cos \lambda_1, \dots, \cos \lambda_n)$ ✓
- ☐ B. $\cos(A) = P^{-1} \cos(D) P$, where $\cos(D) = \text{diag}(\cos \lambda_1, \dots, \cos \lambda_n)$
- ☐ C. $\cos(A) = \cos(P) \cos(D) \cos(P^{-1})$
- ☐ D. $\cos(A) = \text{diag}(\cos \lambda_1, \dots, \cos \lambda_n)$ always, even if A is not diagonal

Your answer is correct.

The correct answer is: $\cos(A) = P \cos(D) P^{-1}$, where $\cos(D) = \text{diag}(\cos \lambda_1, \dots, \cos \lambda_n)$

Question 11

Correct

Mark 1.00 out of 1.00

If $A = \begin{bmatrix} 3 & -3 & -4 \\ 0 & 2 & 2 \\ 0 & 0 & k \end{bmatrix}$, and $|A| = -6$.

i) Find the eigenvalues of the matrices A , $2A$, A^T , A^3 , A^{-1} .

Which option is completely correct?

☒ A. $k = -1$ ✓

Eigenvalues:

- $A: \{-1, 2, 3\}$
- $2A: \{-2, 4, 6\}$
- $A^T: \{-1, 2, 3\}$
- $A^3: \{-1, 8, 27\}$
- $A^{-1}: \{-1, \frac{1}{2}, \frac{1}{3}\}$

☐ B. $k = -1$

Eigenvalues:

- $A: \{-1, 2, 3\}$
- $2A: \{-1, 4, 6\}$
- $A^T: \{-1, 2, 3\}$
- $A^3: \{-1, 8, 27\}$
- $A^{-1}: \{-1, \frac{1}{2}, \frac{1}{3}\}$

☐ C. $k = 1$

Eigenvalues:

- $A: \{1, 2, 3\}$
- $2A: \{2, 4, 6\}$
- $A^T: \{1, 2, 3\}$
- $A^3: \{1, 8, 27\}$
- $A^{-1}: \{1, \frac{1}{2}, \frac{1}{3}\}$

☐ D. $k = -2$

Eigenvalues:

- $A: \{-2, 2, 3\}$
- $2A: \{-4, 4, 6\}$
- $A^T: \{-2, 2, 3\}$
- $A^3: \{-8, 8, 27\}$
- $A^{-1}: \{-\frac{1}{2}, \frac{1}{2}, \frac{1}{3}\}$

Your answer is correct.

The correct answer is:

$k = -1$
Eigenvalues:

- $A: \{-1, 2, 3\}$
- $2A: \{-2, 4, 6\}$
- $A^T: \{-1, 2, 3\}$

- $A^3: \{-1, 8, 27\}$
- $A^{-1}: \{-1, \frac{1}{2}, \frac{1}{3}\}$

Question 12

Correct

Mark 2.00 out of 2.00

If $A = \begin{bmatrix} 3 & -3 & -4 \\ 0 & 2 & 2 \\ 0 & 0 & k \end{bmatrix}$, and $|A| = -6$.

ii) Find the eigenvectors of the matrix A .

Which option lists three vectors that are all eigenvectors of A ?

- ☐ A. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$
- ☐ B. $\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$
- ☐ C. $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$
- ☒ D. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -4 \\ 6 \end{bmatrix} \right\}$ ✓

Your answer is correct.

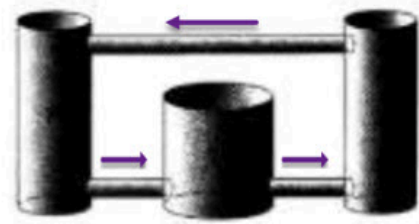
The correct answer is: $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -4 \\ 6 \end{bmatrix} \right\}$

Question 13

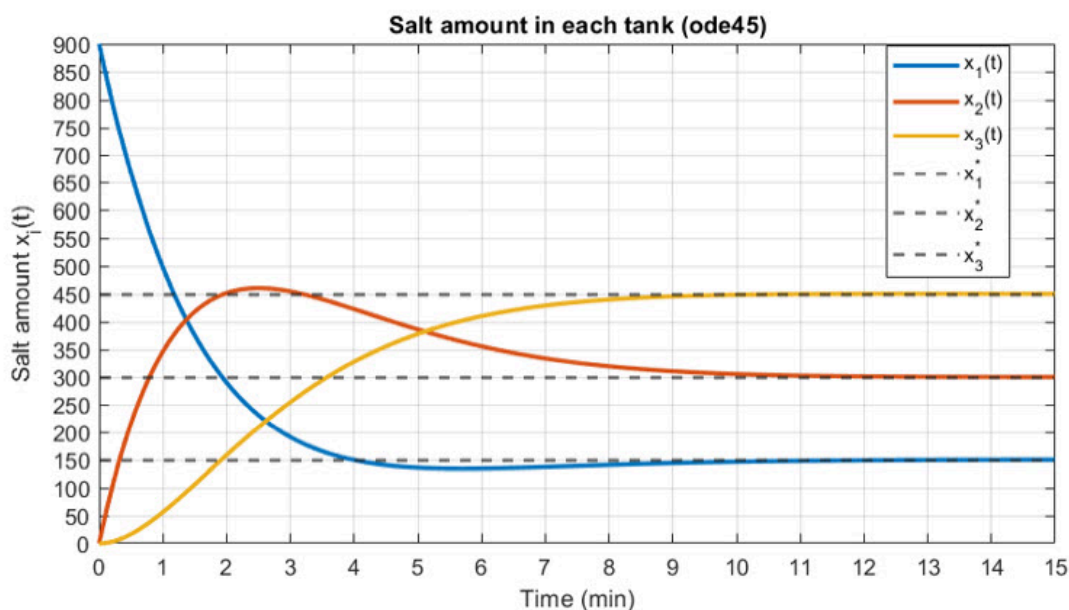
Complete

Mark 2.00 out of 2.00

A closed loop consists of three well-mixed tanks connected by one-way pipes. Brine flows from Tank 1 → Tank 2 → Tank 3 → back to Tank 1 with a constant flow rate r . The tanks have volumes V_1, V_2, V_3 . Let $x_1(t), x_2(t), x_3(t)$ denote the amount of salt in Tanks 1, 2, and 3 (respectively) at time t (minutes). You are provided with:



1. The process description + diagram of the three-tank loop.
2. The time-response plot of $x_1(t), x_2(t), x_3(t)$ (salt amount in each tank versus time), as shown.



Required to answer

- a) Using the plot, and what happens during the first stage of the process (near $t = 0$):
 - Which tank initially has the largest salt amount and why? With stating the initial amount of Salt.
 - Which tank initially has zero or very small salt amount and why?
- b) Using the plot and the “well-mixed, constant-flow” assumptions, deduce a Relationship between tank volumes i.e. $(V_1 : V_2 : V_3)$.
- c) The Amount of Salt in Each Tank after 180 second and 720 second.

a)

- tank 1 $x_1(t)$ has the largest amount of salt as the blue curve starts with amount 900 ode45 while other tanks equal zero, at initial conditions --> $x_1(0) = 900$
- tank 2 and 3 are equal to 0 ode45 as they both the red and yellow curves start from zero @ initial conditions --> $x_2(0) = 0, x_3(0) = 0$

b) at steady state the first tank reduced from 900 ode45 to 150 ode45 and the second tank increased from 0 to 300 ode45 and the third one increased from 0 to 450 ode45 so, the relationship between the 3 tanks will be **150 : 300 : 450** which equal to **1:2:3** so, we can say that the volume of second tank is twice the first one and the volume of the third tank equal to the sum of both tanks 1 & 2.

c)

- after 180 sec. --> 3 min.
 - $x_1(3) = 200$ ode45

- $v_2(3) = 450$ ode45
- $v_3(3) = 250$ ode45

- after 720 sec. --> 12 min.
 - $v_1(12) = 150$ ode45
 - $v_2(12) = 300$ ode45
 - $v_3(12) = 450$ ode45

Comment: